

From Restaurant Growth to an Exact Partition Atlas: Ewens Laws, Block-Count Fluctuations, and Poisson Limits

HCS-C353 source-local reconstruction

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Abstract

We carry the one-parameter Chinese-restaurant insertion rule through a complete finite and asymptotic partition atlas. A labelled-partition induction proves exchangeability and the Ewens sampling formula; exact counting gives the full occupancy-vector law. The total number of blocks is factored into independent Bernoulli innovations, yielding its exact probability generating polynomial, almost-sure logarithmic law, and variance-normalized central limit theorem. A separate mixed falling-factorial identity proves joint independent-Poisson limits for every fixed collection of block sizes. A 5,238-row exact certificate checks finite coefficients and conventions but does not replace the asymptotic proofs. No priority or arithmetic-target claim is made.

Revision certificate. Round-one innovation factorization, strong-law, and CLT closure.

1 Growth rule and complete theorem

Fix $\theta > 0$. Set $\Pi_1 = \{\{1\}\}$. Given a partition Π_n of $[n] = \{1, \dots, n\}$, customer $n+1$ starts a singleton block with probability $\theta/(\theta+n)$, or joins an existing block B with probability $|B|/(\theta+n)$. Let $K_n = |\Pi_n|$ and let $C_j(n)$ be the number of blocks of size j . Write $(a)^{\overline{n}} = a(a+1) \cdots (a+n-1)$ and use $(x)_r = x(x-1) \cdots (x-r+1)$.

Theorem 1 (Finite-to-asymptotic partition atlas). *For the preceding process:*

(i) Π_n is exchangeable. A particular labelled partition $\{B_1, \dots, B_k\}$ has probability

$$\mathbb{P}(\Pi_n = \{B_1, \dots, B_k\}) = \frac{\theta^k}{(\theta)^{\overline{n}}} \prod_{i=1}^k (|B_i| - 1)!. \quad (1)$$

(ii) If $c_1, \dots, c_n \geq 0$ and $\sum_j jc_j = n$, then

$$\mathbb{P}(C_j(n) = c_j, 1 \leq j \leq n) = \frac{n!}{(\theta)^{\overline{n}}} \prod_{j=1}^n \frac{(\theta/j)^{c_j}}{c_j!}. \quad (2)$$

(iii) There are independent Bernoulli variables I_i with $\mathbb{P}(I_i = 1) = \theta/(\theta+i-1)$ such that $K_n = \sum_{i=1}^n I_i$. Consequently

$$\mathbb{E}z^{K_n} = \frac{(\theta z)^{\overline{n}}}{(\theta)^{\overline{n}}}. \quad (3)$$

(iv) As $n \rightarrow \infty$,

$$\frac{K_n}{\log n} \rightarrow \theta \quad \text{almost surely}, \quad \frac{K_n - \mathbb{E}K_n}{\sqrt{\text{Var } K_n}} \Rightarrow N(0, 1). \quad (4)$$

(v) For every fixed m ,

$$(C_1(n), \dots, C_m(n)) \Rightarrow (Z_1, \dots, Z_m), \quad Z_j \text{ independent}, \quad Z_j \sim \text{Poisson}(\theta/j). \quad (5)$$

2 Labelled induction and the occupancy law

Formula (1) holds at $n = 1$. Assume it at n . If $n+1$ creates a new block, multiplication by $\theta/(\theta+n)$ adds one power of θ and the next rising-factorial denominator. If it joins a block of size b , multiplication by $b/(\theta+n)$ changes $(b-1)!$ to $b!$. Every labelled partition at time $n+1$ has a unique predecessor obtained by deleting label $n+1$, so induction proves (1) without multiplicity ambiguity. Its dependence only on block sizes proves exchangeability.

For a count vector $c = (c_1, \dots, c_n)$, the number of labelled set partitions with those counts is

$$\frac{n!}{\prod_{j=1}^n (j!)^{c_j} c_j!}. \quad (6)$$

Multiplying (6) by (1), with $\prod_i (|B_i| - 1)! = \prod_j ((j - 1)!)^{c_j}$, gives (2). Summing (2) also yields

$$(\theta)^{\overline{n}} = \sum_{k=1}^n c(n, k) \theta^k,$$

where $c(n, k)$ is the unsigned Stirling number of the first kind.

3 Independent innovations, strong law, and CLT

Let I_i indicate that customer i opens a new block; $I_1 = 1$. Crucially,

$$\mathbb{P}(I_i = 1 \mid \Pi_1, \dots, \Pi_{i-1}) = p_i := \frac{\theta}{\theta + i - 1} \quad (7)$$

is deterministic. Iterated conditioning therefore factors every finite joint event in the I_i , proving mutual independence rather than merely uncorrelatedness. Since $K_n = \sum I_i$, multiplying their Bernoulli PGFs gives (3). In particular

$$\begin{aligned} a_n &:= \mathbb{E} K_n = \sum_{i=1}^n p_i = \theta \log n + O(1), \\ v_n &:= \text{Var } K_n = \sum_{i=1}^n p_i(1 - p_i) = \theta \log n + O(1). \end{aligned} \quad (8)$$

The second estimates use $p_i = \theta/i + O(i^{-2})$ and $\sum_i p_i^2 < \infty$.

Set $X_i = I_i - p_i$. Independence and

$$\sum_{i \geq 2} \frac{\text{Var } X_i}{(\log i)^2} < \infty$$

imply, by the Kolmogorov convergence theorem, that $\sum_{i \geq 2} X_i / \log i$ converges almost surely. Kronecker's lemma gives $\sum_{i \leq n} X_i / \log n \rightarrow 0$ almost surely. Combining this with (8) proves the first limit in (4).

For the second, $v_n \rightarrow \infty$ while $|X_i| \leq 1$. Thus for every $\varepsilon > 0$, the Lindeberg indicators $\mathbf{1}_{\{|X_i| > \varepsilon \sqrt{v_n}\}}$ vanish identically once n is large. The Lindeberg–Feller theorem proves the centered, variance-normalized limit in (4).